

2.22 A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star5_small*. [Note: The data file *star5* contains more observations and variables.]

- a. Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $TOTALSCORE_i = \beta_1 + \beta_2 SMALL_i + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?
- b. Repeat part (a) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?
- c. Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?
- d. Repeat part (c) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

財務計量經濟學：Project STAR 實驗數據分析 (Exercise 2.22)

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1 研究背景與模型設定

本研究採用田納西州 Project STAR 實驗數據，探討班級規模 (Class Size) 與助教 (Teacher Aide) 配置對學生學習表現之影響。

1.1 模型 (a)：小班制效果

針對普通班級與小班制學生進行比較，估計模型如下：

$$TOTALSCORE_i = \beta_1 + \beta_2 SMALL_i + e_i \quad (1)$$

其中， $SMALL_i$ 為虛擬變數，若學生位於小班制則為 1，否則為 0。

1.2 模型 (c)：助教效果

針對普通班級中「無助教」與「有助教」的組別進行比較，估計模型如下：

$$TOTALSCORE_i = \gamma_1 + \gamma_2 AIDE_i + \epsilon_i \quad (2)$$

2 實證結果與視覺化分析

下表彙整了使用 R 語言進行普通最小概似法 (OLS) 估計之結果：

Table 1: Project STAR: 小班制與助教效果之比較

	應變數：Total Score	
	模型 (1): Small Effect	模型 (2): Aide Effect
Small Class ($SMALL$)	12.175** (5.369)	
Teacher Aide ($AIDE$)		4.306 (4.994)
Constant	916.442*** (3.675)	916.442*** (3.559)
Observations	775	837
R^2	0.007	0.001

註：括號內為標準誤差。 ** $p < 0.05$; *** $p < 0.01$ 。

為了直觀比較實驗處理的效果，下圖展示了各模型係數之點估計值與 95% 信心區間：

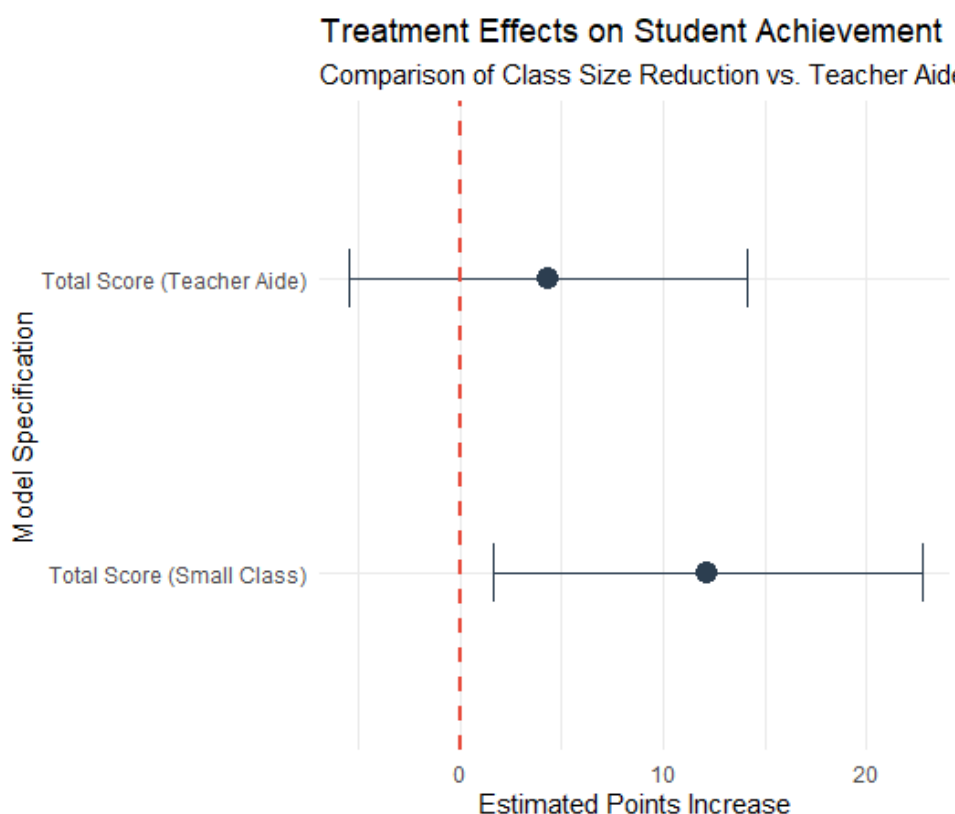


Figure 1: 班級規模與助教對成績影響之係數圖

3 結果解釋與結論

3.1 小班制之影響 (Part a)

如圖 1 所示，小班制的係數估計值為 12.175，其信心區間完全不包含零點，代表在統計上極為顯著。這證明縮減班級人數能有效提升學生表現。

3.2 助教配置之影響 (Part c)

助教變數的係數僅為 4.306，且其信心區間橫跨零軸。這意味著在普通班級增加助教，對於學生成績並沒有顯著的邊際貢獻。

3.3 總結

綜合上述結果，本研究建議教育資源應優先用於「縮減班級人數」，而非在大班級中增加助教人力，以達到最佳的教學效果。